

MICRO-ARCHITECTURE DOCUMENT

Shift Register System (SISO, SIPO, PISO, PIPO)

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1. Introduction

This document describes the micro-architecture for a unified shift-register module supporting four modes:

SISO, SIPO, PISO, and PIPO. Serial data valid and serial output valid signals are integrated for reliable communication.

2. Module Overview

Inputs:

- clk — common system clock
- rst_n — active-low reset
- one_hot[3:0] — selects shift register mode
- siso_serial_in — serial input for SISO
- sipo_serial_in — serial input for SIPO
- piso_parallel_in[3:0] — parallel data for PISO load
- piso_load — load/shift control
- pipo_parallel_in[3:0] — parallel data input
- serial_data_valid — indicates valid serial input

Outputs:

- siso_serial_out — serial output of SISO
- sipo_parallel_out[3:0] — parallel output of SIPO
- piso_serial_out — serial output of PISO
- pipo_parallel_out[3:0] — parallel output of PIPO
- serial_out_valid — output bit is valid

3. One-Hot Mode Selection

one_hot[0] → SISO

one_hot[1] → SIPO

one_hot[2] → PISO

one_hot[3] → PIPO

Only one active at a time.

4. Serial Valid Signal Handling

serial_data_valid controls when serial inputs are consumed:

- In SISO, serial input updates output only when valid.
- In SIPO, shift occurs only when input is valid.
- In PISO shift mode, the register shifts only when valid.

serial_out_valid asserts:

- In SISO whenever output bit is updated.
- In PISO during shift cycles.

5. SISO Micro■Architecture

No internal registers required.

Operation:

If mode = SISO and serial_data_valid = 1:

siso_serial_out ← siso_serial_in

serial_out_valid ← 1 during update.

6. SIPO Micro■Architecture

Internal register: 4-bit shift register.

On serial_data_valid:

shift left and insert serial input.

serial_out_valid = 0 always.

7. PISO Micro■Architecture

Internal register: shift_reg_piso[3:0]

Two operations:

- Load: piso_load = 1 → load parallel data.
- Shift: piso_load = 0 and serial_data_valid = 1 → shift left.

Output bit: MSB of shift_reg_piso.

serial_out_valid = 1 during shifting.

8. PIPO Micro■Architecture

Simple combinational transfer:

If mode = PIPO:

pipo_parallel_out = pipo_parallel_in

serial_out_valid = 0 always.

9. Reset Behavior

On rst_n = 0:

All outputs and internal registers reset to 0.

10. Datapath Summary

SISO:

serial_in → output flop → serial_out

SIPO:

serial_in → 4-bit shift register → parallel_out

PISO:

parallel_in → shift_reg → serial_out

PIPO:

parallel_in → parallel_out

11. Control Path Summary

- one_hot determines active mode.
- serial_data_valid gates serial operations.
- piso_load controls PISO load/shift.
- serial_out_valid communicates output validity.

12. Conclusion

This micro-architecture provides an efficient unified design with strong validity signaling, suitable for synthesis and verification.